

Coordination science reads synchrony as connection. Capoeira is about knowing when to **break it** — and when to **restore it**.

This is a dynamic interplay of playfulness, pressure, resistance, freedom and cunning — staying oneself, and transforming, while engaging the world. The small roda (circle) rehearses the big roda of life.

BEYOND STILLNESS

Contemplative research centres on solitary, meditation-based practice. Capoeira Angola — an Afro-Brazilian art born of resistance — cultivates contemplative capacities differently: attentiveness, vigilance, discernment amid vulnerability, trained inside a contested, musical, communal game.

ATTUNEMENT, THEN RUPTURE

Two players tune in — reading weight, timing, the other's breath — and settle into a shared rhythm. Then one breaks it: a kick, a feint, a sudden drop of level. The rupture is not a failure of coordination but the move itself — it desynchronises the pair and reorganises the exchange into something new. Capoeira lives in this making and breaking of synchrony.

SUBTLE SHIFTS, REAL CHANGE

Each sync-desync transition retunes the game from second to second — and, over a life of practice, the players themselves: how they attend, resist, and stay themselves while engaging another. These subtle shifts are what we set out to catch — where they happen, in the movement and in the telling.

WHO PLAYS, AND HOW

15–20 experienced *angoleiros* (≥5 years of practice) in Salvador, Bahia, recruited through established academies; a senior *Mestre* guards the ritual and ethical frame. Within-subjects, four conditions along the presence axis:

- ① solo *ginga* to a steady instrumental rhythm — the baseline attractor;
- ② dyadic jogo with no circle around;
- ③ the same dyads inside a small roda — 13–18 peers, a pre-selected song, the *Mestre* holding the tempo;
- ④ a naturalistic roda — song, tempo and intensity unconstrained, the *Mestre* leading as in authentic practice.

Sessions run in a real capoeira space; sensors are exchanged between players without breaking the roda's flow.



THREE PERSPECTIVES, ONE JOGO

- **1st person:** immediate experience reports + video-cued micro-phenomenological interviews — "here I felt trapped", "here I broke free".
- **2nd person:** Mestre's culturally grounded commentary.
- **3rd person:** high-frequency IMUs + markerless pose estimation from synchronized multi-camera video.

TOWARD A KINEPHENOMENOLOGY

Varela's neurophenomenology held first-person experience and neural dynamics in mutual constraint — each disciplining the other. I attempt the same reciprocity with movement as the substrate: micro-phenomenological accounts and the dynamics of the jogo — coupling, relative phase, recurrence (RQA) — guide and correct one another.

WHAT WE ASK

Do modes of agency have distinct dynamical signatures? Do felt moments of sync and rupture appear as measurable phase transitions? Does the roda — attention, song, the Mestre — amplify the coupling between experience and movement?

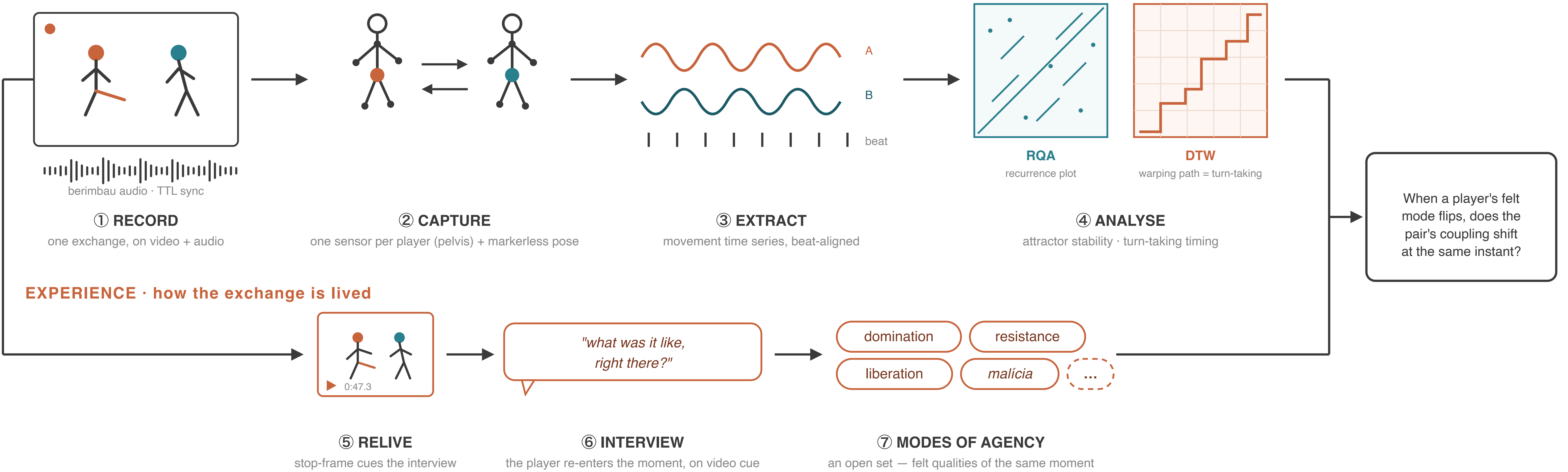
FOR DISCUSSION: THE RESEARCHER IN THE RODA

I study capoeira as a practitioner. Where does this researcher-participant stance sit between first-second- and third-person methods? Can embodied skill serve as an instrument rather than a bias? **Come argue with me.**

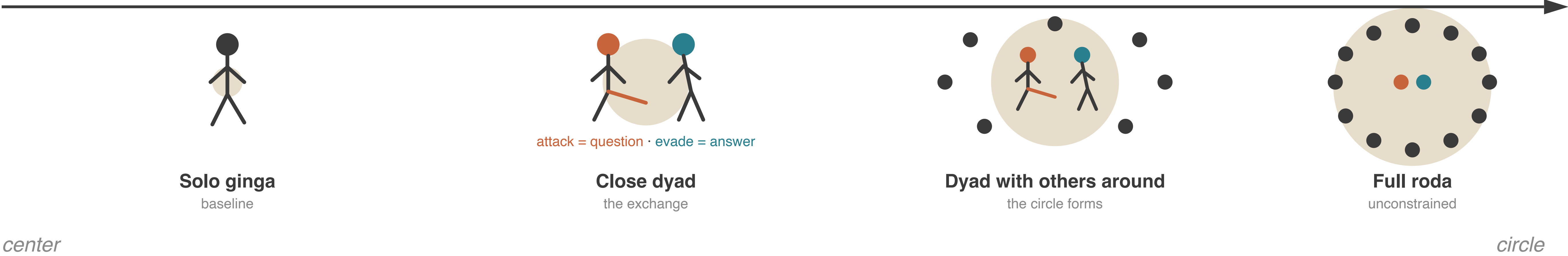


► Scan QR code to view the jogo

MOVEMENT · dynamics (3rd-person)



PRESENCE OF THE OTHER



The presence gradient: from solo ginga to the full, unconstrained roda.